main.c 10 th

```c

#include <stdio.h>
#include <string.h>

int main(void) {
    char data[100], divisor[30], temp[140], codeword[140];
    int n, m, i, j, error = 0;

    printf("Enter data bits: ");
    scanf("%99s", data);
    printf("Enter generator bits (e.g. 1011): ");
    scanf("%29s", divisor);

    n = (int)strlen(data);
    m = (int)strlen(divisor);

```

input

```
Enter data bits (0s and 1s only): 1101
Original data : 1101
Stuffed frame : 01111110110101111110


...Program finished with exit code 0
Press ENTER to exit console.
```

main.c

```c
#include <stdio.h>
#include <string.h>
int main(void) {
 char data[200], stuffed[420];
 int i, j = 0, count = 0;
 printf("Enter data bits (0s and 1s only): ");
 scanf("%199s", data);
 /* Opening flag: 01111110 */
 strcpy(stuffed, "01111110");
 j = strlen(stuffed);
for (i = 0; data[i] != '\0'; i++) {
    if (data[i] != '0' && data[i] != '1') {
        printf("Invalid input: enter only 0 and 1.\n");
        return 1;
        }
 stuffed[j++] = data[i];
```

input

```
Enter data bits (0s and 1s only): 111110111111
Original data : 111110111111
Stuffed frame : 0111111011111001111101011111110


...Program finished with exit code 0
Press ENTER to exit console.
```

input

```
Enter total number of frames: 7
Enter window size: 3

Sliding Window Protocol Simulation

Sender window: frames 0 to 2
Sender: Sending frame 0
Sender: Sending frame 1
Sender: Sending frame 2
Receiver: Frame 0 received
Receiver: ACK 0 sent
Receiver: Frame 1 received
Receiver: ACK 1 sent
Receiver: Frame 2 received
Receiver: ACK 2 sent
Sender: Window moved forward

Sender window: frames 3 to 5
Sender: Sending frame 3
Sender: Sending frame 4
Sender: Sending frame 5
Receiver: Frame 3 received
Receiver: ACK 3 sent
Receiver: Frame 4 received
Receiver: ACK 4 sent
Receiver: Frame 5 received
Receiver: ACK 5 sent
Sender: Window moved forward

Sender window: frames 6 to 6
Sender: Sending frame 6
Receiver: Frame 6 received
Receiver: ACK 6 sent
Sender: Window moved forward
```